



KAHLERT SCHOOL OF COMPUTING

THE UNIVERSITY OF UTAH

MiniWeb

SPARK Lab

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Ideastorming

Motivation

- **Definition:** Existing web agent benchmarks have the same constituting ***sub-tasks called macros***. For example - *purchasing a product on a shopping site*.
- **Redundancy:** When compiling the macros from many different popular benchmarks, we find to be very limited and repetitive.
- **Evaluation:** How do these agents perform when the *macros are treated as an evaluation axis instead of treating domains*?
- **Idea:** We want to have both macro- and domain-level diversity.
 - **Infra:** To create such a hyper-scalable *containerized* web environment, we need better web infrastructure.
 - **Infra:** We strip all redundant backend components from web envs, and keep frontend.
- **Method:** We can identify the macros these agents struggle at and focus on them better (by using better training methods).
- Can we evaluate generalization in a better way on unseen macros instead of domains?
- Macro-level continual learning + error analysis.
- Cross-benchmark Transfer -> Training in MiniWeb does better in WebArena and vice versa.

Macros and Domains

Formal Definition of Macros

- Short horizon
- Without any detailed planning
- Can not be broken down into simpler sub-tasks without defining the actual action (chain of atomic actions)
- Naturally occurs in a website
- We can have multiple types of Macros

E.g., logging it into an account on a website, purchasing an item on an e-commerce site.

Compilation of Macros

1. Compile macros from the existing web benchmarks
2. Brainstorm for novel macros (also do this with LLMs)
3. Survey people and for novel macros

Compilation of Domains (nature of websites)

1. Existing benchmarks
2. Brainstorm for novel domains (also do this with LLMs)
3. Survey people for domains that they use

Domains, Tasks, and Evaluation

How are we building the domains?

- Automatic Domain Creation using Coding Agents (<https://arxiv.org/pdf/2603.10505>)/WA-Inf/Gym
- Manual Verification of those domains

How are we making tasks?

- **Defn.** Tasks are essentially what will be fed to the agent for evaluation.
- Ask AI to combine the macros and domains then manually review. (Tasks can have 1+ macros)
- Ensure we have task splits -
 - Macro-level - Average Success Rate for each macro
 - Macro-length - Could be correlated to complexity?
 - Domain-level - Average Success Rate in each web domain
 - Macro-type Analysis
 - Seen-unseen across all the previous ones (keep some of the tasks unseen)

Evaluation (Reward Model)

- Define evaluation for each task
- We can have a set of evaluators, and we can use multiple for each task if needed.
 - E.g., list matching, exact matching, fuzzy matching, VLM judge, etc.
- The LLM can pick which evaluators they wanna use for each task

Experiment Design

- Train on MiniWeb and evaluate on MiniWeb and analyze across macros and domains.
- **Baselines:**
 - Zero-shot (text, image)
 - Behavior Cloning (SFT baseline)
 - GRPO (RL baseline)
 - Skill baseline
- **Exp:** Can we manually building skills based on the macros and use those skills instead?
- **Cross-benchmark Generalization:**
 - Agents trained on MiniWeb would do better on novel benchmarks than agents trained on other env.
 - We train agents on WebArena, WebArena-Infinity, WebGym, Insta, and on MiniWeb
 - We evaluate both agents on a new benchmark, like Online Mind2Web
- **Mini-Web as an environment for Mid-training (seems tricky):**
 - We train agents on MiniWeb before fine-tuning it on a benchmark and see that it improves performance vs. fine-tuning on that benchmark solely.
- **Macro-level Continual Learning (we can skip this if it's too much to cover)**
- **Analysis:**
 - Which macros and domains agents struggle with more?
 - How does performance scale with the number of macros and domains?
 - **Error Analysis:** Which macros agents still struggle with after training?

Concerns

- Since we are taking macros from existing benchmarks, are the tasks essentially *seen*?
 - Ans: Taking macros is not the same as taking tasks.
- Maybe less robust to backend variations (auth flows, session expiry, server-side validation)
 - We can maybe introduce that backend variations artificially in miniweb
 - Isn't that essentially a macro?
 - Modify that domain so that error step can be evaluated as a part of the task (being its own macro)

Timeline

1. Phase-1 (Benchmark Creation): (Milestone - End of June)

- ~~a. Compile the list of benchmarks~~
- b. Compile macros - Schedule a weekly 1 hour slot, maybe get another member -> Approach Reaz!
- c. Compile the domains
- d. Create the domains using automatic pipelines and verify
- e. Compile tasks using automatic pipelines and verify
- f. Add the evaluation for each task

2. Phase-2 (Method and Experimenting): (Milestone - End of July)

- a. Create the baseline methods
- b. Trained on miniweb and get the results
- c. Set other environments for cross-benchmark generalization and mid-training evaluation

3. Phase-3 Analysis and Paper Writing (~August)

Thank You!

Steal this taxonomy diagram

